

Summer 2016-2017 Final Exam

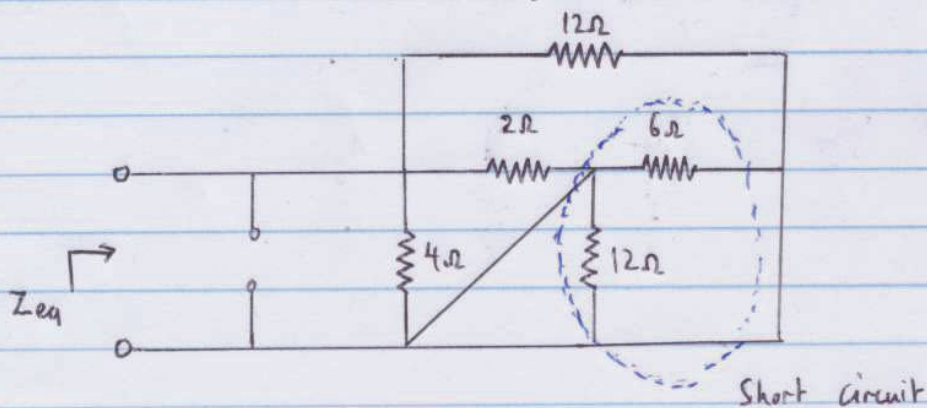
Question 1

i)

a) Under DC conditions

$0.5F$ capacitor $\Rightarrow$ Open circuit

$4H$ inductor $\Rightarrow$ Short circuit



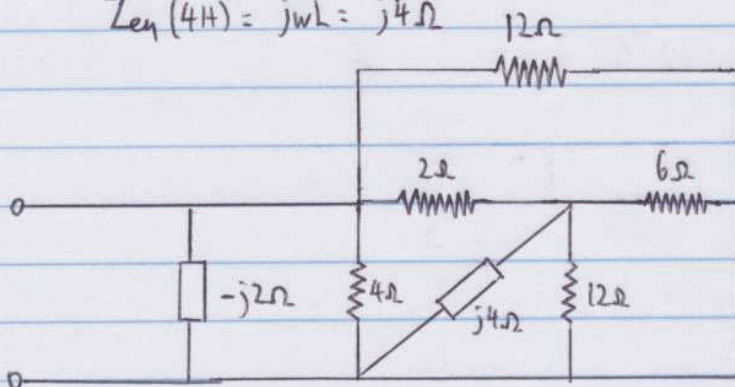
$$Z_{eq} = 12 \parallel 2 \parallel 4$$

$$= \left(\frac{1}{12} + \frac{1}{2} + \frac{1}{4} \right)^{-1} = 1.2 \Omega$$

b) $\omega = 1 \text{ rad/s}$

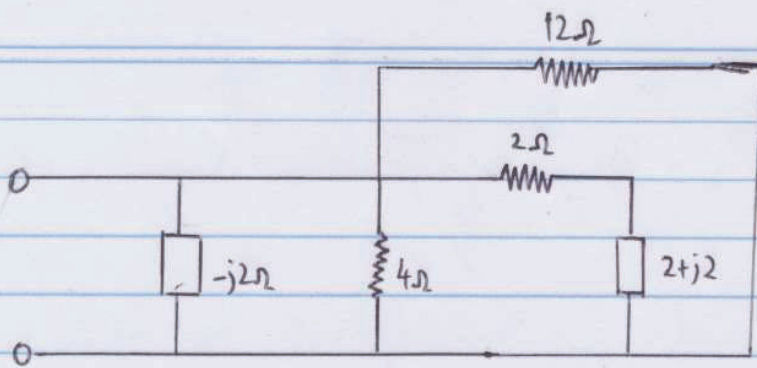
$$Z_{eq}(0.5F) = \frac{-j}{\omega C} = \frac{-j}{0.5} = -j2 \Omega$$

$$Z_{eq}(4H) = j\omega L = j4 \Omega$$



$$6 \parallel 12 \parallel j4 = \left(\frac{1}{6} + \frac{1}{12} + \frac{1}{j4} \right)^{-1}$$

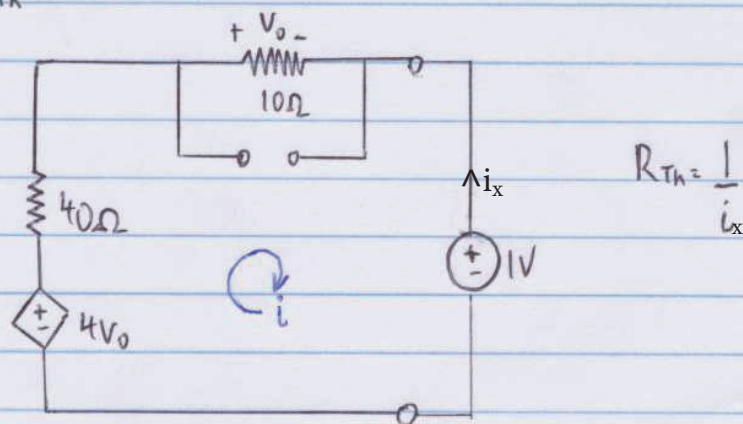
$$= (2 + j2) \Omega$$



$$Z_{eq} = -j2 // 4 // (4+j2) // 12$$

$$= \left(\frac{1}{-j2} + \frac{1}{4} + \frac{1}{4+j2} + \frac{1}{12} \right)^{-1} = (1.2 - j0.9) \Omega$$

ii) Finding R_{Th}



KVL in mesh:

$$-4V_0 + 40i + V_0 + 1 = 0$$

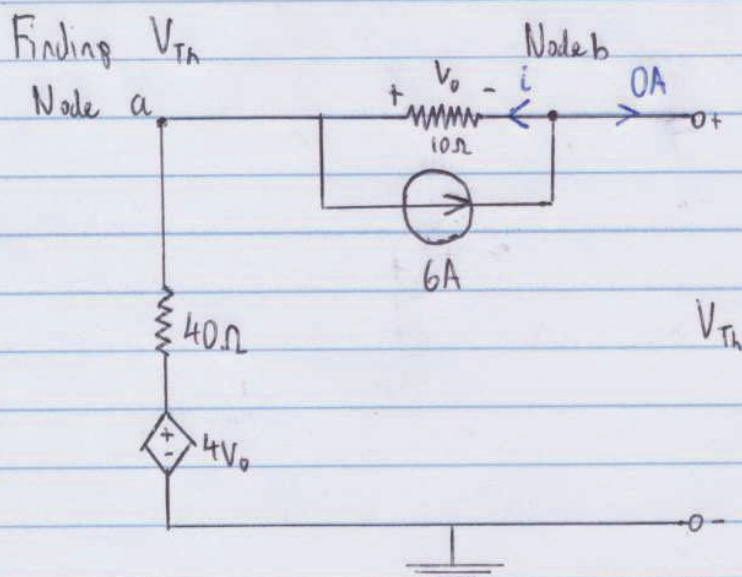
$$\text{But } V_0 = 10i$$

$$40i - 3(10i) = -1$$

$$10i = -1$$

$$i = -0.1 \text{ A} \Rightarrow i_x = 0.1 \text{ A}$$

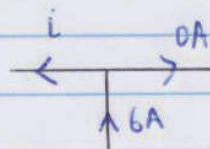
$$\therefore R_{Th} = 10 \Omega$$



KCL at node b:

$$i + 0 = 6$$

$$\therefore i = 6A$$



Also: $6 = \frac{V_b - V_a}{10} \Rightarrow V_a - V_b = -60$

KCL at node a:

$$\frac{V_a - (4 \times -60)}{40} + 6 + \frac{V_a - V_b}{10} = 0$$

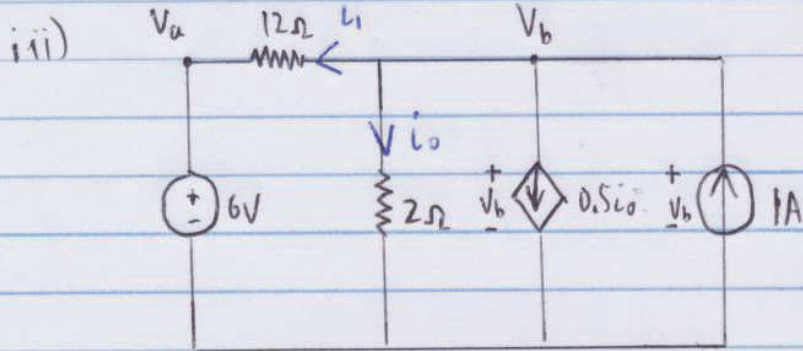
$$V_a + 240 + 240 + 4V_a - 4V_b = 0$$

$$5V_a - 4V_b = -480$$

$$\begin{pmatrix} V_a & V_b \\ 1 & -1 \\ 5 & -4 \end{pmatrix} \begin{vmatrix} -60 \\ -480 \end{vmatrix} \quad R_2 = R_2 - 5R_1$$

$$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{vmatrix} -60 \\ -180 \end{vmatrix} \quad \therefore V_b = -180V$$

$$\therefore V_{Th} = V_b - 0 = -180V$$



KCL at node b:

$$\frac{V_b - 6}{12} + \frac{V_b}{2} + \frac{1}{2} \left(\frac{V_b}{2} \right) = 1$$

$$V_b - 6 + 6V_b + 3V_b = 12$$

$$10V_b = 18$$

$$V_b = 1.8V$$

Finding i_1 :

$$i_1 = \frac{V_b - V_a}{12} = \frac{1.8 - 6}{12} = -0.35A$$

$$P_{6V} = -0.35 \times 6 = 2.1W \text{ supplied}$$

$$P_{1A} = 1.8 \times -1 = 1.8W \text{ supplied}$$

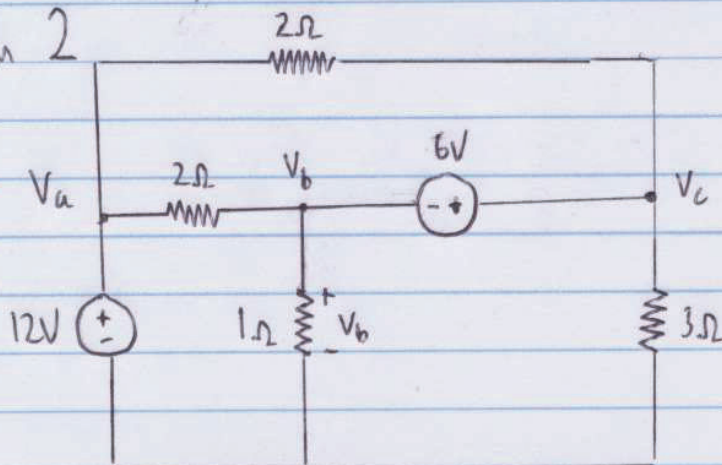
$$P_{0.5i_1} = \frac{1}{2} \left(\frac{1.8}{2} \right) \times 1.8 = 0.81W \text{ absorbed}$$

Explanation:

By sign convention, positive power absorbs and negative power supplies.

Question 2

i)



$V_a = 12V$ due to source

Supernode:

$$V_c - V_b = 6V \quad (1)$$

KCL at supernode:

$$\frac{V_b - 12}{2} + \frac{V_b}{1} + \frac{V_c - 12}{2} + \frac{V_c}{3} = 0$$

$$3V_b - 36 + 6V_b + 3V_c - 36 + 2V_c = 0$$

$$9V_b + 5V_c = 72 \quad (2)$$

$$\begin{pmatrix} V_b & V_c \\ -1 & 1 \\ 9 & 5 \end{pmatrix} \begin{vmatrix} 6 \\ 72 \end{vmatrix} \quad R_2 = R_2 + 9R_1$$

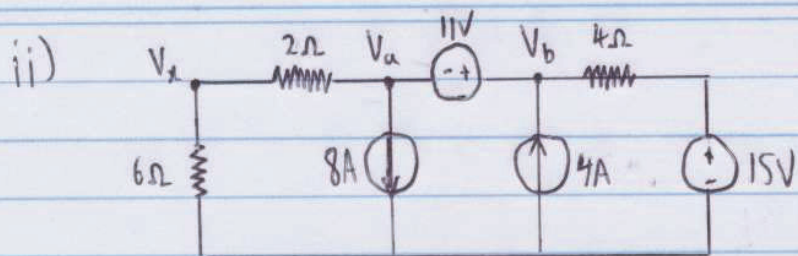
$$\begin{pmatrix} -1 & 1 \\ 0 & 14 \end{pmatrix} \begin{vmatrix} 6 \\ 126 \end{vmatrix}$$

$$\therefore V_c = \frac{126}{14} = 9V$$

$$V_b = -(6 - 9) = 3V$$

$$\therefore V_{1\Omega} = 3V$$

$$P_{3\Omega} = \frac{V_c^2}{R} = \frac{9^2}{3} = 27W \text{ absorbed}$$



KCL at node x:

$$\frac{V_x}{6} + \frac{V_x - V_a}{2} = 0$$

$$4V_x - 3V_a = 0 \quad (1)$$

Supernode between nodes a and b:

$$V_b - V_a = 11V \Rightarrow V_a - V_b = -11 \quad (2)$$

KCL at supernode:

$$\frac{V_a - V_x}{2} + 8 + \frac{V_b - 15}{4} = 4$$

$$2V_a - 2V_x + 32 + V_b - 15 = 16$$

$$-2V_x + 2V_a + V_b = -1 \quad (3)$$

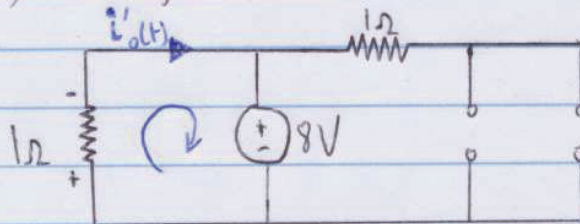
$$\begin{pmatrix} V_x & V_a & V_b \\ 4 & -3 & 0 & 0 \\ 0 & 1 & -1 & -11 \\ -2 & 2 & 1 & -1 \end{pmatrix} \quad R_3 = 2R_3 + R_1$$

$$\begin{pmatrix} 4 & -3 & 0 & 0 \\ 0 & 1 & -1 & -11 \\ 0 & 1 & 2 & -2 \end{pmatrix} \quad R_3 = R_3 - R_2$$

$$\begin{pmatrix} 4 & -3 & 0 & 0 \\ 0 & 1 & -1 & -11 \\ 0 & 0 & 3 & 9 \end{pmatrix} \quad \begin{aligned} \therefore V_b &= 3V \\ V_a &= -8V \\ \underline{V_x} &= \underline{-6V} \end{aligned}$$

iii) Must use superposition

Only turning on 8V source on



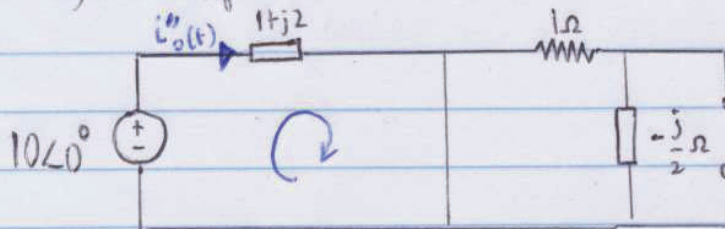
Inductor $\Rightarrow$ Closed circuit
Capacitor $\Rightarrow$ Open circuit

KVL:

$$i'_o + 8 = 0$$

$$i'_o = -8A$$

Only turning on $10\cos(2t)V$ source on



$$\omega = 2 \text{ rad/s}$$

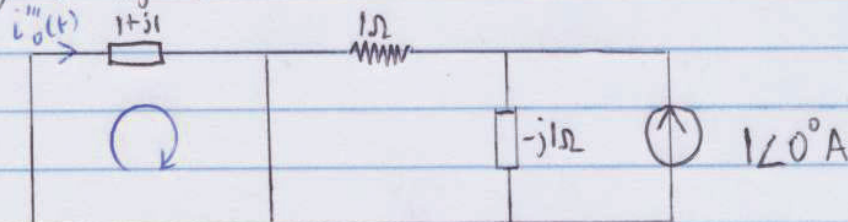
$$\text{Inductor} \Rightarrow j\omega L = j2\Omega$$

$$\text{Capacitor} \Rightarrow \frac{-j}{\omega C} = \frac{-j}{2}\Omega$$

$$\text{KVL: } (1+j2)i''_o - 10 = 0$$

$$i''_o = \frac{10}{1+j2} = 4.472\cos(2t - 63.4^\circ)A$$

Only turning on $1\cos(t)A$ source on



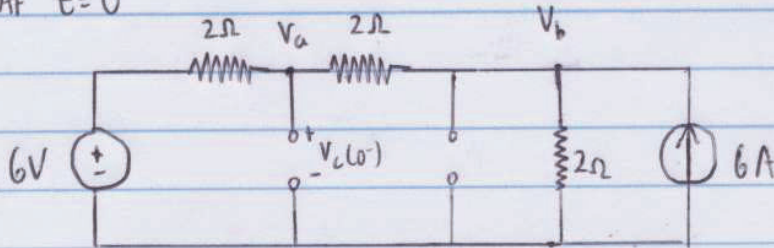
$$i'''_o(t) = 0A \text{ due to short circuit}$$

$\therefore$ By superposition:

$$i_o(t) = [-8 + 4.472\cos(2t - 63.4^\circ)]A$$

Question 3

i) At $t=0^-$



KCL at node a:

$$\frac{V_a - 6}{2} + \frac{V_a - V_b}{2} = 0$$

$$2V_a - V_b = 6 \quad (1)$$

KCL at node b:

$$\frac{V_b}{2} + \frac{V_b - V_a}{2} = 6$$

$$-V_a + 2V_b = 12 \quad (2)$$

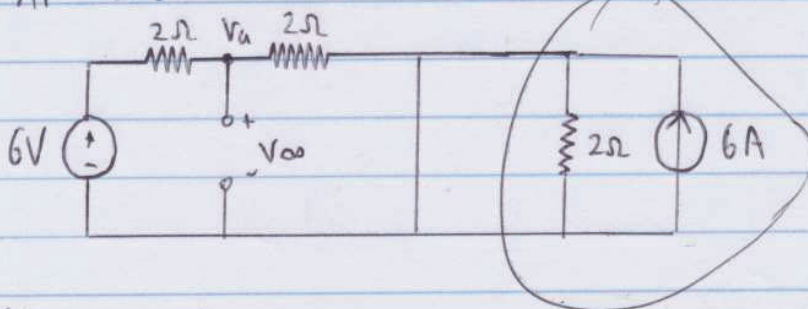
$$(2) \times 2 \Rightarrow -2V_a + 4V_b = 24 \quad (3)$$

$$(1) + (3) \Rightarrow 3V_b = 30$$

$$V_b = 10V$$

$$\therefore V_a = 8V = V_c(0^-)$$

At $t=0^+$



Short circuit

KCL at node a:

$$\frac{V_a - 6}{2} + \frac{V_a}{2} = 0$$

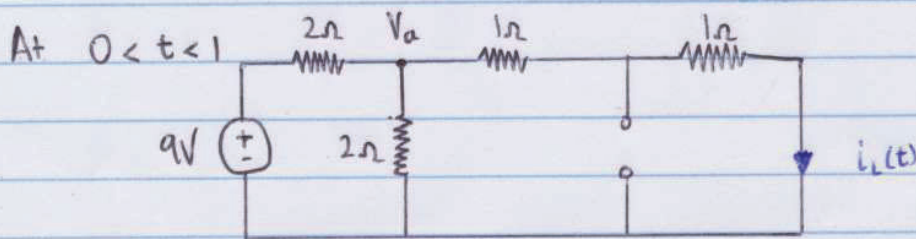
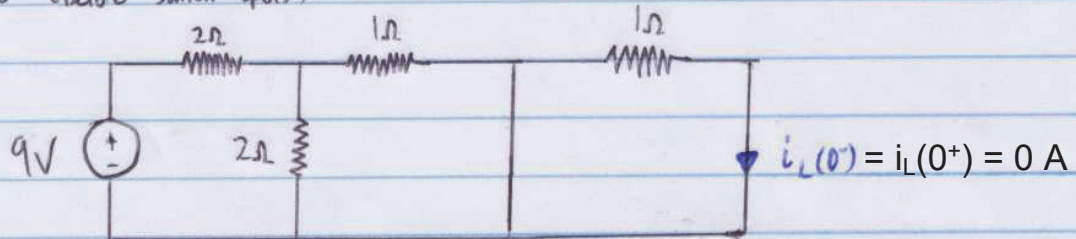
$$R_{Th} = 2 \parallel 2 \Rightarrow \tau = RL$$

$$= 1 \Omega \quad = 10 \text{ ms} = 10 \times 10^{-3}$$

$$V_L(t) = V_{\infty} + (V_L(0^-) - V_{\infty}) e^{-\frac{t}{\tau}} \text{ V}$$

$$= 3 + 5e^{-100t} \text{ V for } t > 0$$

ii) At $t = 0^-$ (Before switch opens)



KCL at node a:

$$\frac{V_a - 9}{2} + \frac{V_a}{2} + \frac{V_a}{2} = 0$$

$$3V_a = 9$$

$$V_a = 3 \text{ V}$$

$$\therefore i_L(t) = \frac{3}{2} \text{ A} = i_L(\text{infinity})$$

Assuming that the circuit remains in this condition for a long time

$$R_{Th} = 2 + 2 \parallel 2 = 3 \Omega$$

$$\tau = \frac{L}{R_{Th}} = \frac{4}{3} \text{ s}$$

$$i_L(t) = 1.5 - 1.5e^{-\frac{3}{4}t} \text{ A for } 0 < t < 1$$

At $t > 1$ (Circuit is the same as first one in question)

$$i_{\infty} = 0 \text{ A}$$

$$i(1) = 1.5(1 - e^{-0.75}) = 0.79 \text{ A}$$

$$R_{Th} = 1 \Omega$$

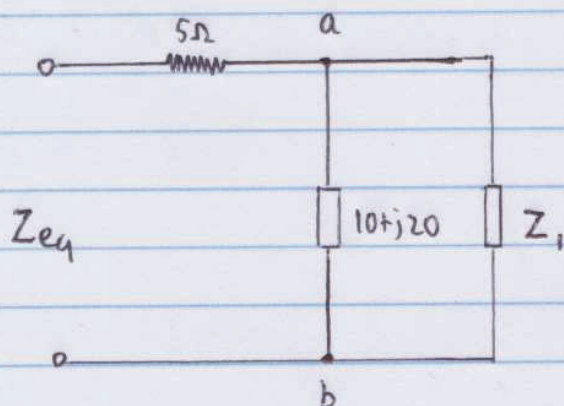
$i(1)$ is used as initial value in the equation instead of $i(0^+)$ since the switch changes at $t = 1$ s.

$$T = \frac{L}{R} = 4 \text{ s}$$

$$i_L(t) = \begin{cases} 0.79 e^{-\frac{(t-1)}{4}} \text{ A} & \text{for } t > 1 \end{cases}$$

$$i_L(t) = \begin{cases} 1.5(1 - e^{-\frac{3}{4}t}) \text{ A} & \text{for } 0 < t < 1 \\ 0.79 e^{-\frac{(t-1)}{4}} \text{ A} & \text{for } t > 1 \end{cases}$$

iii)



$$Z_1 = a + jb$$

$$Z_{eq}(2H) = j(2)(10) = j20 \Omega$$

$$Z_{eq} = 5 + Z_1 // 10 + j20$$

For Z_{eq} to be purely resistive

$$\text{Im}(Z_1 // 10 + j20) = 0$$

$$Z_1 // 10 + j20 = \frac{(a + jb)(10 + j20)}{(a + j10) + j(b + j20)} = \frac{(10a - 20b) + j(20a + 10b)}{(a + 10) + j(b + 20)} = \frac{r_1 \angle \theta_1}{r_2 \angle \theta_2}$$

If $\theta_1 = \theta_2$ the circuit will be purely resistive:

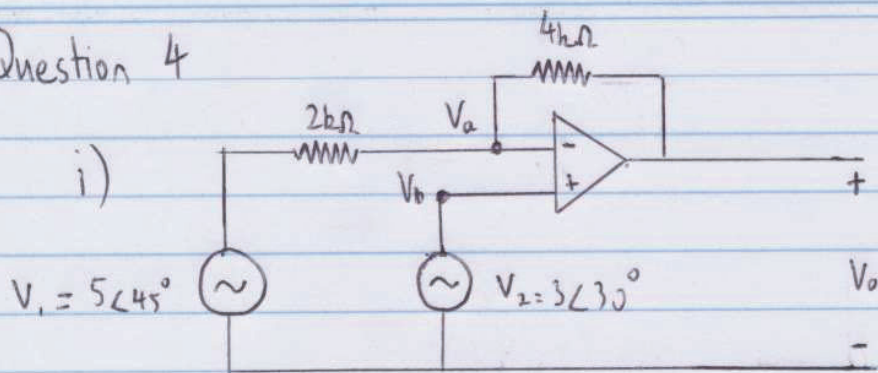
$$\frac{20a + 10b}{10a - 20b} = \frac{b + 20}{a + 10}$$

$$20a^2 + 20b^2 + 500b = 0 \Rightarrow a^2 + b^2 + 25b = 0 \Rightarrow a^2 + b(b + 25) = 0$$

$$b = -25 \quad a = 0$$

$$\therefore Z_1 = -j25 = \frac{-j}{\frac{1}{25}} = \frac{-j}{100} \Rightarrow C = \frac{1}{250} = 4 \text{ mF}$$

Question 4



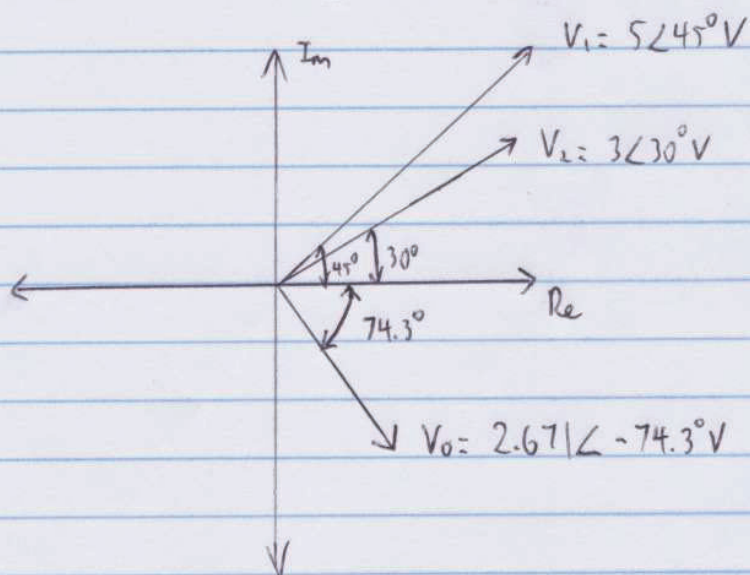
$$V_a - V_b = 3\angle 30^\circ$$

KCL at node a:

$$\frac{3\angle 30^\circ - 5\angle 45^\circ}{2,000} + \frac{3\angle 30^\circ - V_0}{4,000} = 0$$

$$\therefore V_0 = 4,000 \left(\frac{3\angle 30^\circ - 5\angle 45^\circ}{2,000} + \frac{3\angle 30^\circ}{4,000} \right)$$

$$= 2.671\angle -74.3^\circ \text{ V}$$

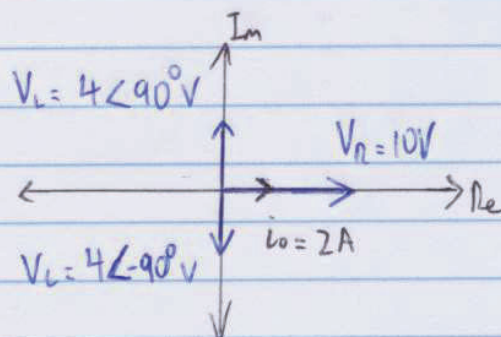


ii) $\omega = 1 \text{ rad/s}$

$$Z_{eq}(2H) = j2\Omega$$

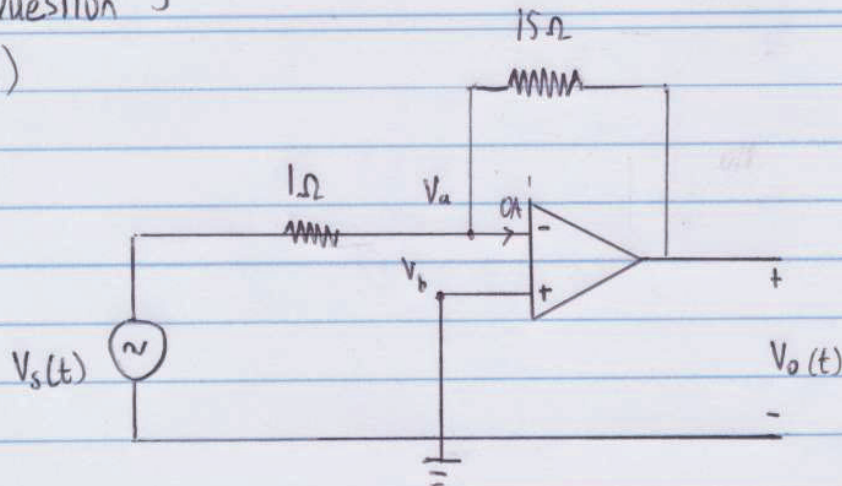
$$Z_{eq}(0.5F) = -j2\Omega$$

$$I_0 = \frac{10\angle 0^\circ}{5} = 2 \text{ A}$$



Question 5

i)



$$V_b = 0V$$

$$V_b = V_a \text{ (Principle of Op amp)}$$

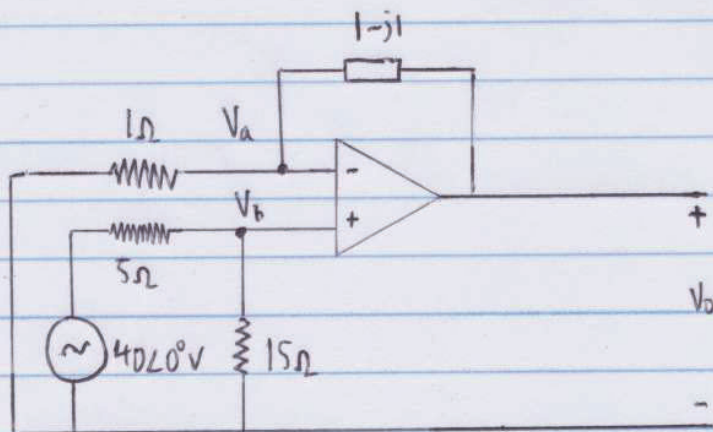
KCL at node a

$$\frac{0 - V_s(t)}{1} + \frac{0 - V_o(t)}{15} = 0$$

$$-V_s(t) = \frac{V_o(t)}{15}$$

$$\therefore V_o(t) = -15V_s(t)$$

ii)



KCL at node b:

$$\frac{V_b - 40 \angle 0^\circ}{5} + \frac{V_b}{15} = 0$$

$$3V_b - 120 + V_b = 0$$

$$V_b = 30 \text{ V} = V_a \text{ (Op amp principle)}$$

KCL at node a:

$$\frac{V_a}{1} + \frac{V_a - V_o}{1-j} = 0$$

$$30 + \frac{30}{1-j} = \frac{V_o}{1-j}$$

$$\therefore V_o = (1-j) \left(30 + \frac{30}{1-j} \right)$$

$$= 60 - j30 = 67.082 \angle -26.57^\circ \text{ V}$$

IGNORE THIS QUESTION

iii) a) $Z = \overline{A+B} + \overline{B+C}$

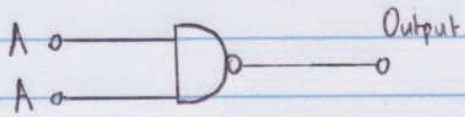
$$\begin{aligned} &= \overline{A+B} \cdot \overline{B+C} = (A+\bar{B}) \cdot (B+C) \\ &= A.B + A.C + \bar{B}.B + \bar{B}.C \\ &= A.B + A.C + \bar{B}.C \end{aligned}$$

b)

A	B	C	A.B	A.C	$\bar{B}.C$	Z
0	0	0	0	0	0	0
0	0	1	0	0	1	1
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	0	0	0	0
1	0	1	0	1	1	1
1	1	0	1	0	0	1
1	1	1	1	1	0	1

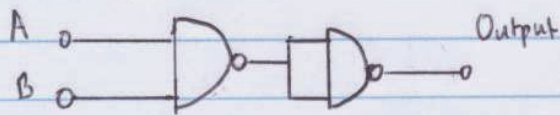
c)

i) NOT gate



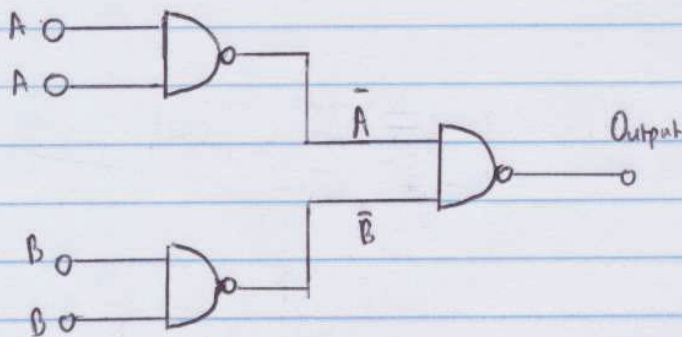
$$\text{Output} = \overline{A \cdot A} = \overline{A + \overline{A}} = \overline{A}$$

ii) AND gate



$$\text{Output} = \overline{\overline{A \cdot B}} = A \cdot B$$

iii) OR gate



$$\text{Output} = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A} + \overline{B}} = A + B$$